

Chapter 3

Data Link Layer

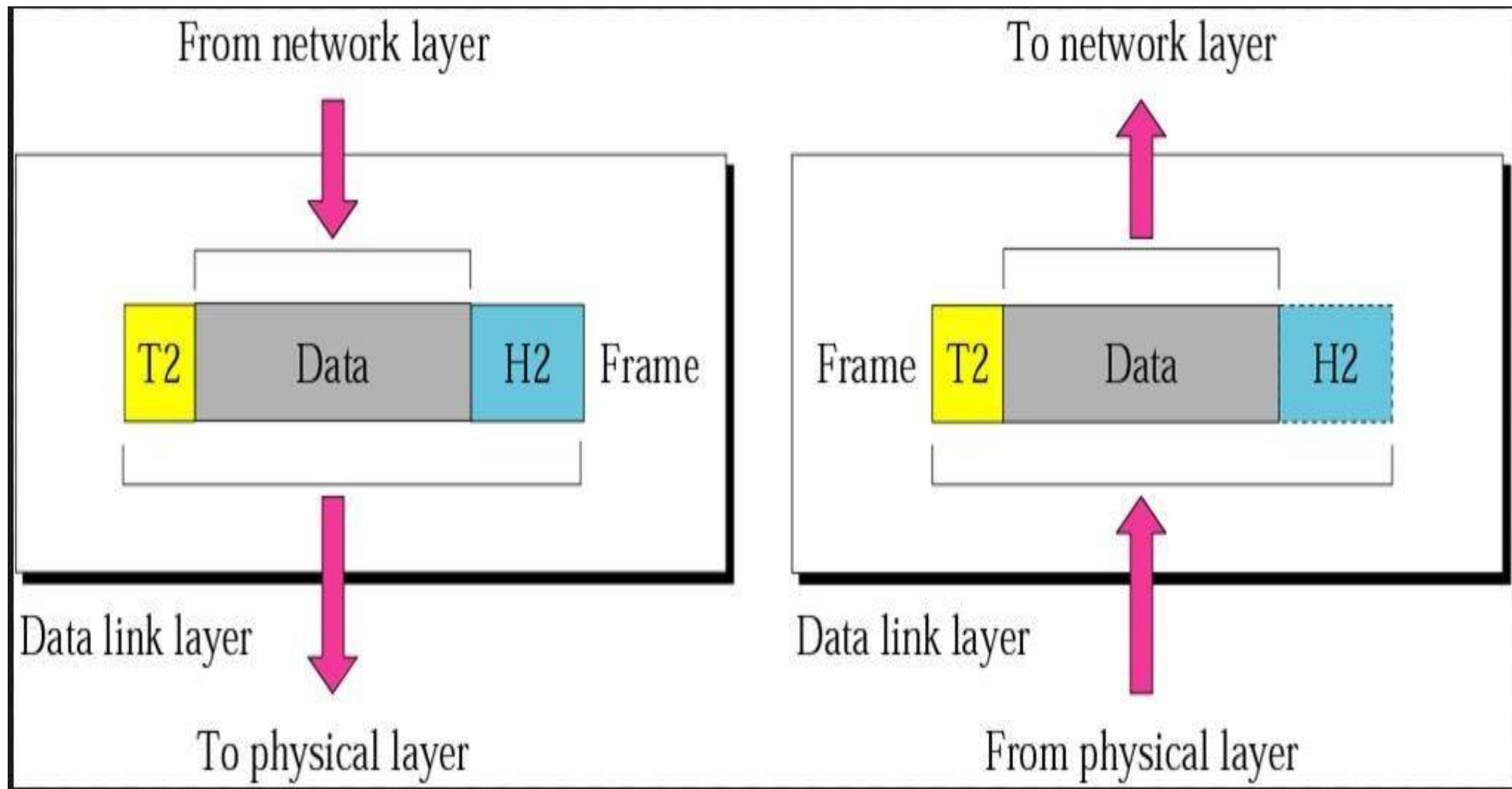
Introduction

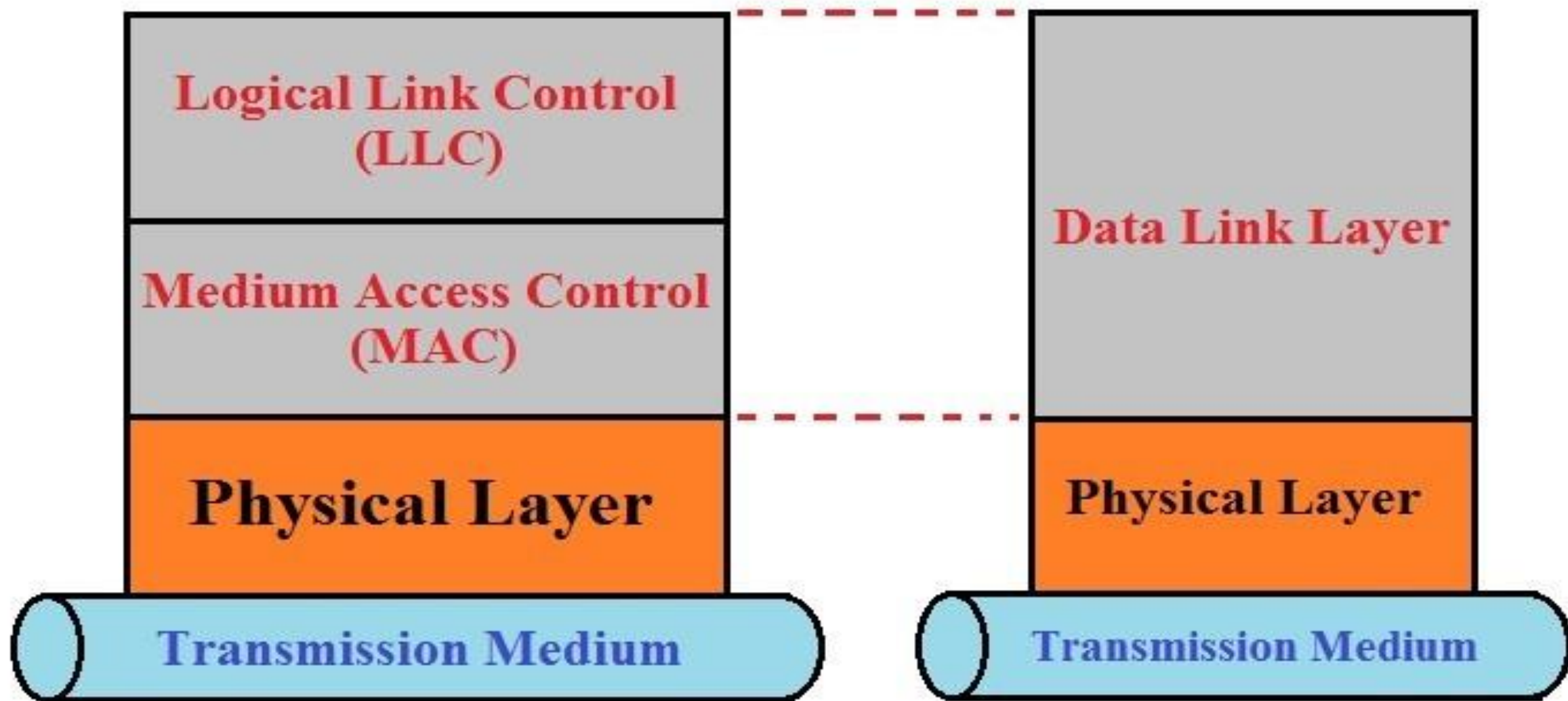
The Data Link Layer is the second layer of the OSI Layered Model.

The data link layer has two sub-layers:

Logical Link Control(LLC): It deals with protocols, flow-control, and error control.

Media Access Control(MAC): It deals with the actual control of media.





3.1 Functions of Data Link Layer

For effective data communication between two nodes, the data link layer has to carry out several specific functions as follows:

1. **Framing:** The data link layer divides the stream of bits received from the network layer into manageable data units called frames.
2. **Physical addressing:** The data link layer adds a header(MAC address) to the packet, which becomes a frame to define the sender and/or receiver of the frame.

If the frame is intended for a system outside the sender's network, the receiver address is the address of the device that connects the network to the next one.

Functions of Data Link Layer

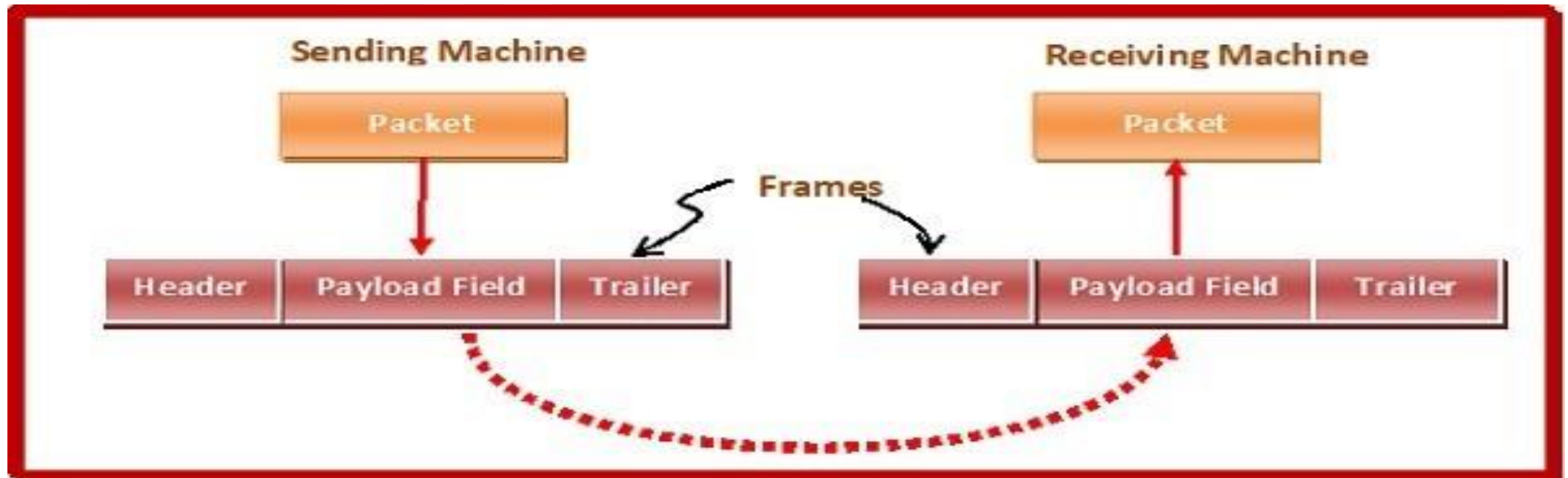
- 3. Flow control:** If the rate at which the receiver absorbs the data is less than the rate at which data are produced in the sender, the data link layer imposes a flow control mechanism to avoid overwhelming the receiver.
- 4. Error control:** The data link layer adds reliability to the physical layer by adding mechanisms to detect and retransmit damaged or lost frames. Error control is normally achieved through a trailer added to the end of the frame.
- 5. Access control:** When two or more devices are connected to the same link, data link layer protocols are necessary to determine which device has control over the link at any given time.
- 6. Synchronization:** When data frames are sent on the link, both machines must be synchronized to transfer.

3.2 Data Link Control: Framing, Flow and Error Control

1. Framing

- The data link layer takes the packet from the network layer and encapsulates it into frames.
- If the frame size becomes too large, then the packet may be divided into small sized frames which makes flow control and error control more efficient.
- Then, it sends each frame bit-by-bit on the hardware.
- At receiver's end, data link layer picks up signals from hardware and assembles them into frames.

Framing





A frame has the following parts:

- a) Frame Header – It contains the source and the destination addresses of the frame.
- b) Payload Field – It contains the message to be delivered.
- c) Trailer – It contains the error detection and error correction bits.
- d) Flag – It marks the beginning and end of the frame(delimiter).

Types of Framing

Framing can be of two types, fixed sized framing and variable sized framing.

a) Fixed-Sized Framing

Here, the frame size is fixed, so the frame length acts as a delimiter. Consequently, it does not require additional boundary bits to identify the frame's start and end.

b) Variable – Sized Framing

- Here, the size of each frame to be transmitted may be different. So, additional mechanisms are kept to mark the end of one frame and the beginning of the next frame.

Two ways to define frame delimiters in variable-sized framing are

Length Field – Here, a length field determines the size of the frame.

End Delimiter – Here, a pattern is used as a delimiter to determine the size of the frame. If the pattern occurs in the message, then two approaches are used –

Byte-Stuffing – A byte is stuffed in the message to differentiate from the delimiter. This is also called character-oriented framing.

Bit-Stuffing– A pattern of bits of arbitrary length is stuffed in the message to differentiate it from the delimiter. This is also called bit-oriented framing.

2. Flow Control:

It is a set of procedures that tells the sender how much data can transmit before the data overwhelms the receiver.

The receiving device has limited speed and limited memory to store the data.

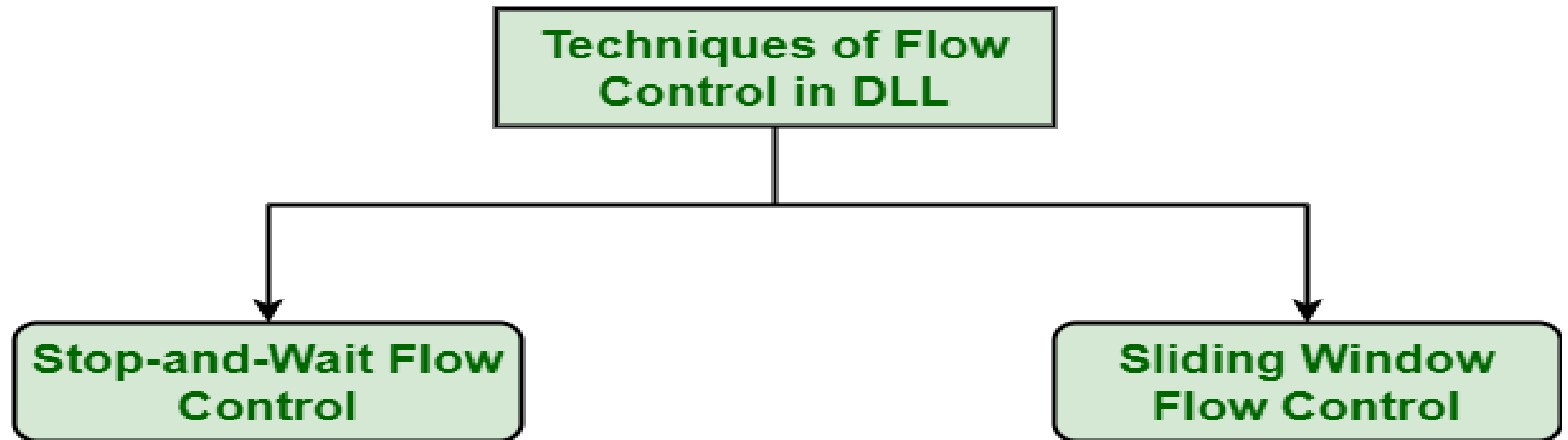
Therefore, the receiving device must be able to inform the sending device to stop the transmission temporarily before the limits are reached.

Two methods have been developed to control the flow of data:

Stop-and-Wait

Sliding Window

Flow Control



a) Stop and Wait:

- This protocol involves the following transitions –
The sender sends a frame and waits for acknowledgment.
Once the receiver receives the frame, it sends an acknowledgment frame back to the sender.
On receiving the acknowledgment frame, the sender understands that the receiver is ready to accept the next frame. So it sends the next frame in the queue.

b)Sliding Window:

This protocol improves the efficiency of the stop-and-wait protocol by allowing multiple frames to be transmitted before receiving an acknowledgment.

Both the sender and the receiver have finite-sized buffers called windows.

The sender and the receiver agree upon the number of frames to be sent based on the buffer size.

The sender sends multiple frames in a sequence, without waiting for acknowledgment.

When its sending window is filled, it waits for acknowledgment.

On receiving acknowledgment, it advances the window and transmits the next frames, according to the number of acknowledgments received.

3. Error Control:

Error control in the data link layer is the process of detecting and correcting data frames that have been corrupted or lost during transmission.

In case of lost or corrupted frames, the receiver does not receive the correct data frame, and the sender is ignorant about the loss.

Data link layer follows a technique to detect a transit error or a frame is lost. The process is called Automatic Repeat Request (ARQ).

Error Control:

The error control mechanism in the data link layer involves the following phases –

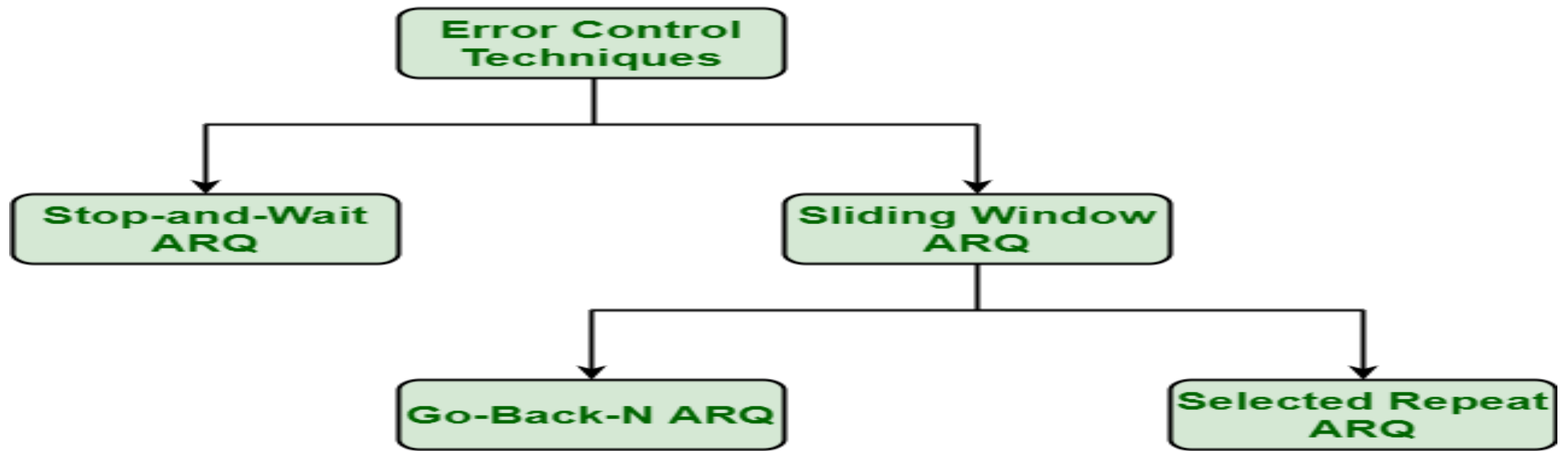
Detection of Error – Transmission error, if any, is detected by either the sender or the receiver.

Acknowledgment – Acknowledgment may be positive or negative.

Positive ACK – On receiving a correct frame, the receiver sends a positive acknowledgment.

Negative ACK – On receiving a damaged frame, the receiver sends a negative acknowledgment back to the sender.

Retransmission – The sender maintains a clock and sets a timeout period. If an acknowledgment of a data frame previously transmitted does not arrive before the timeout, or a negative acknowledgment is received, the sender retransmits the frame



Error Control Techniques

There are three main techniques for error control –

Stop and Wait ARQ

Go-Back-N ARQ

c) Selective Repeat ARQ

a)Stop and Wait ARQ:

- This protocol involves the following transitions –
 - A timeout counter is maintained by the sender, which is started when a frame is sent.
 - If the sender receives acknowledgment of the sent frame within time, the sender is confirmed about successful delivery of the frame. It then transmits the next frame in queue.
 - If the sender does not receive the acknowledgment within time, the sender assumes that either the frame or its acknowledgment is lost in transit. It then retransmits the frame.
 - If the sender receives a negative acknowledgment, the sender retransmits the frame.

b) Go-Back-N ARQ:

The working principle of this protocol is – that the sender has buffers called the sending window.

The sender sends multiple frames based on the acknowledgment of the previous ones.

The receiver receives frames one by one. It keeps track of incoming frames' sequence numbers and sends the corresponding acknowledgment frames.

After the sender has sent all the frames in the window, it checks up to what sequence number it has received a positive acknowledgment.

If the sender has received positive acknowledgment for all the frames, it sends the next set of frames.

If the sender receives N-ACK or has not received any ACK for a particular frame, it retransmits all the frames after which it does not receive any ACK

c) Selective Repeat ARQ

Both the sender and the receiver have buffers called the sending window and the receiving window, respectively

The sender sends multiple frames based on the sending window size without receiving the acknowledgment of the previous ones.

The receiver also receives multiple frames within the receiving window size.

The receiver keeps track of incoming frames' sequence numbers and buffers the frames in memory.

It sends ACK for all successfully received frames and sends N-ACK for only frames that are missing or damaged.

The sender in this case sends only the packet for which N-ACK is received.

In reliable full-duplex data transmission, the technique of hooking up acknowledgments onto outgoing data frames is called piggybacking

3.3 Error Detection and Correction

There are many reasons, such as noise, cross-talk etc., which may help data to get corrupted during transmission.

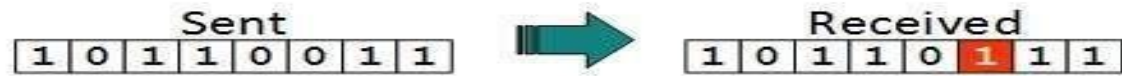
Most of the applications would not function as expected if they received erroneous data.

The data-link layer uses some error control mechanism to ensure that frames (data bit streams) are transmitted with a certain level of accuracy.

Types of Errors

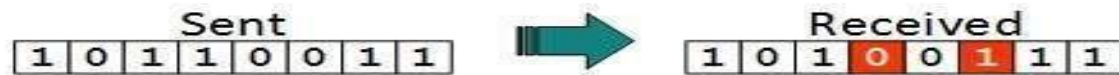
- There may be three types of errors:

1. Single bit error



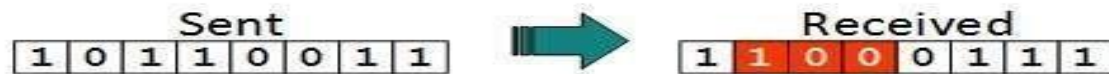
In a frame, there is only one bit, anywhere, though, which is corrupt.

2. Multiple bits error



Frame is received with more than one bits in corrupted state.

3. Burst error



Frame contains more than 1 consecutive bits

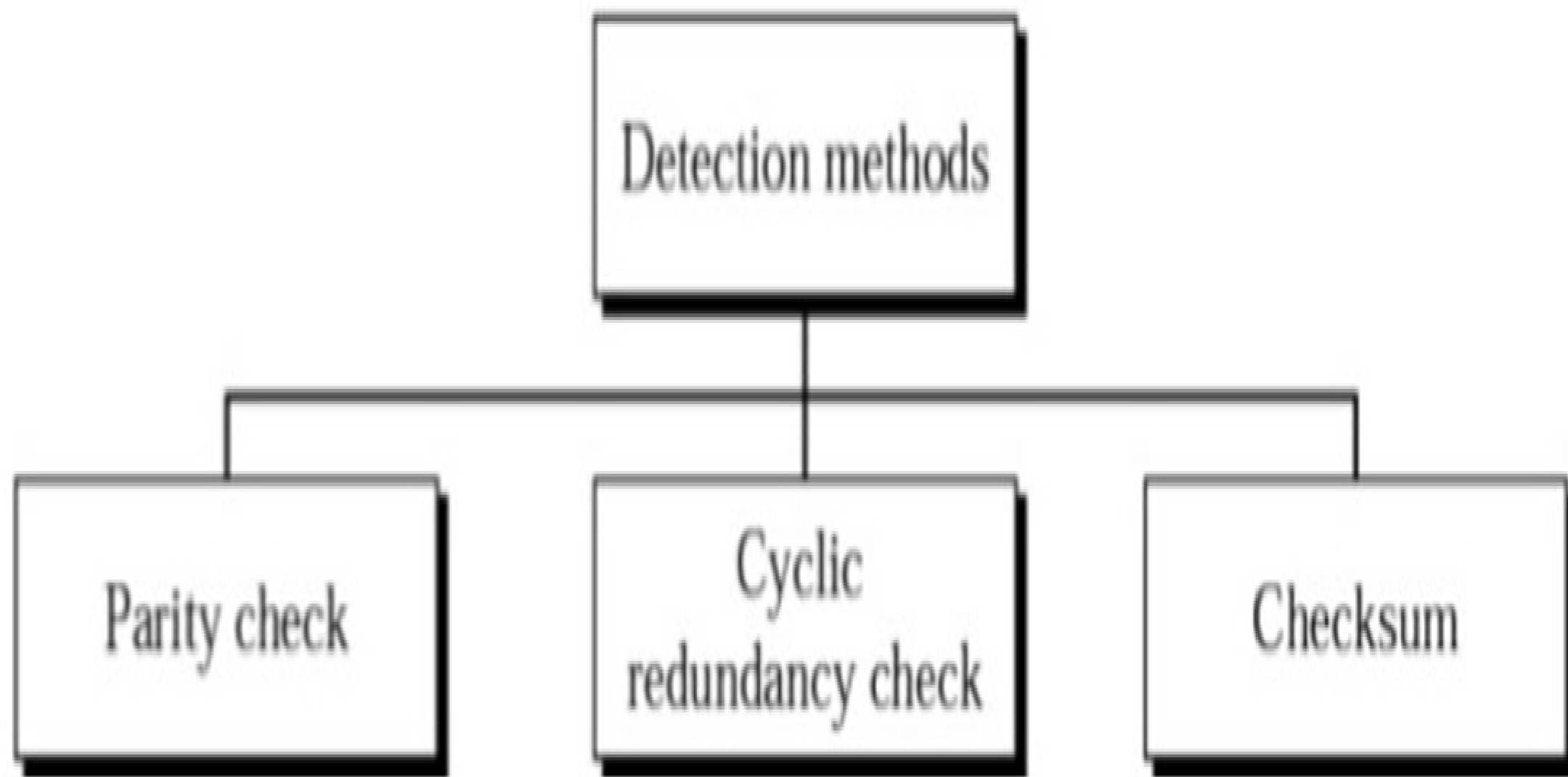
Error Detecting Codes

It is Implemented at Data link layer of OSI Model.

Whenever a message is transmitted, it may get scrambled by noise or data may get corrupted.

To avoid this, we use **error-detecting codes**, which are additional data added to a given digital message to help us detect if any error has occurred during transmission of the message.

Basic approach used for error detection is the use of redundancy bits, where additional bits are added to facilitate the detection of errors.



1. Parity Check:

One extra bit is sent along with the original bits to make the number of 1s either even in case of even parity, or odd in case of odd parity.

It is of the following types:

- a) Simple Parity Check
- b) Two-dimensional Parity check

SENDER

1 0 0 0 1 1



Compute
parity bit



1 0 0 0 1 1 | 1



Transmission
Media



1 0 0 0 1 1 | 1

RECEIVER

Reject Data

N



Even

Y



Accept Data



Compute
parity bit



Two-dimensional Parity check

Parity check bits are calculated for each row, which is equivalent to a simple parity check bit.

Parity check bits are also calculated for all columns, then both are sent along with the data.

At the receiving end these are compared with the parity bits calculated on the received data.

Original Data

10011001	11100010	00100100	10000100
----------	----------	----------	----------

Row parities

1 0 0 1 1 0 0 1	0
1 1 1 0 0 0 1 0	0
0 0 1 0 0 1 0 0	0
1 0 0 0 0 1 0 0	0
1 1 0 1 1 0 1 1	0

Column parities



100110010	111000100	001001000	100001000	110110110
-----------	-----------	-----------	-----------	-----------

Data to be sent

Checksum:

In a checksum error detection scheme, the data is divided into k segments, each of m bits.

In the sender's end, the segments are added to get the sum.

The sum is complemented to get the checksum.

The checksum segment is sent along with the data segments.

At the receiver's end, all received segments are added to get the sum. The sum is complemented.

If the result is zero, the received data is accepted; otherwise discarded.

Original Data

10011001	11100010	00100100	10000100
----------	----------	----------	----------

1

2

3

4

k=4, m=8

Sender

1	10011001
2	11100010
	<u>10111101</u>
	1
	01111100
3	<u>00100100</u>
	10100000
4	<u>10000100</u>
	100100100
	1
Sum:	<u>00100101</u>
Checksum:	11011010

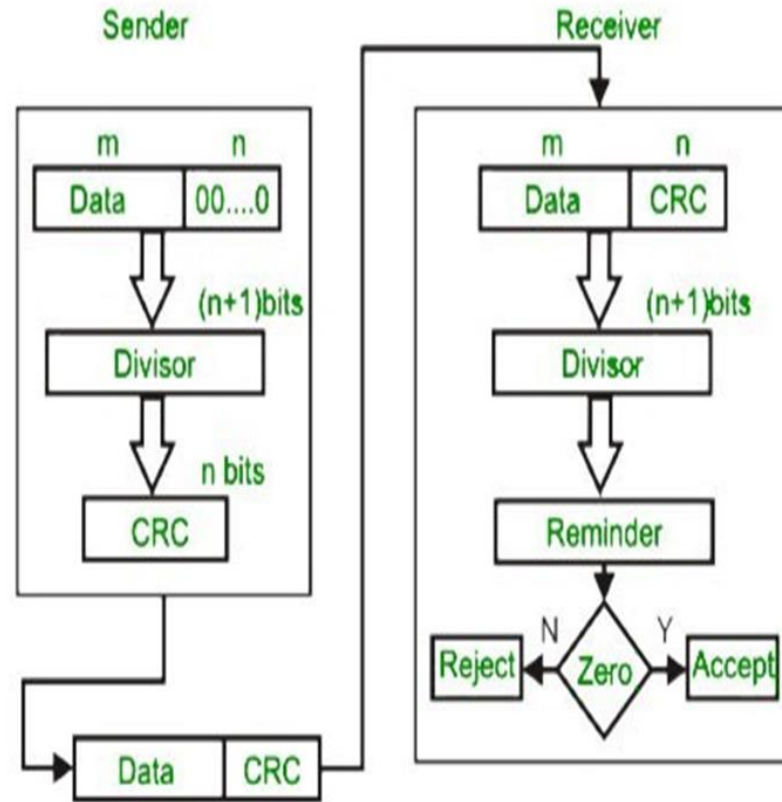
Receiver

1	10011001
2	11100010
	<u>10111101</u>
	1
	01111100
3	<u>00100100</u>
	10100000
4	<u>10000100</u>
	100100100
	1
	00100101
	11011010
Sum:	<u>11111111</u>
Complement:	00000000
Conclusion:	Accept Data

3. Cyclic Redundancy Check (CRC):

- Unlike the checksum scheme, which is based on addition, CRC is based on binary division.
- In CRC, a sequence of redundant bits, called cyclic redundancy check bits, is appended to the end of the data unit so that the resulting data unit becomes exactly divisible by a predetermined binary number.
- At the destination, the incoming data unit is divided by the same number. If at this step there is no remainder, the data unit is assumed to be correct and is therefore accepted.
- A remainder indicates that the data unit has been damaged in transit and therefore must be rejected.

3. Cyclic Redundancy Check (CRC):



original message
1 0 1 0 0 0 0

@ means X-OR

Sender

```
1001 | 10100000000
@1001
-----
0011000000
@1001
-----
01010000
@1001
-----
0011000
@1001
-----
01010
@1001
-----
0011
```

Message to be transmitted

```
1010000000
+ 011
-----
1010000011
```

Generator polynomial
 x^3+1
 $1 \cdot x^3 + 0 \cdot x^2 + 0 \cdot x^1 + 1 \cdot x^0$
CRC generator
1001 4-bit

If CRC generator is of n
bit then append (n-1)
zeros in the end of
original message

```
1001 | 10100000011
@1001
-----
0011000011
@1001
-----
01010011
@1001
-----
0011011
@1001
-----
01001
@1001
-----
0000
```

Receiver

Zero means data is
accepted

Error Correction:

Error correction codes are used to detect and correct the errors when data is transmitted from the sender to the receiver.

Error correction is the additional ability to reconstruct the original, error-free data.

Error correction can be done in two ways:

1. Backward Error Correction: When the receiver detects an error in the data received, it requests that the sender retransmit the data unit.

2. Forward Error Correction: When the receiver detects some error in the data received, it executes an error-correcting code, which helps it to auto-recover and to correct some kinds of errors.

Hamming Code

Hamming code is a set of error-correction codes that can be used to detect and correct errors that can occur when the data is moved or stored from the sender to the receiver.

It is a technique developed by R.W. Hamming for error correction.

Redundant bits:

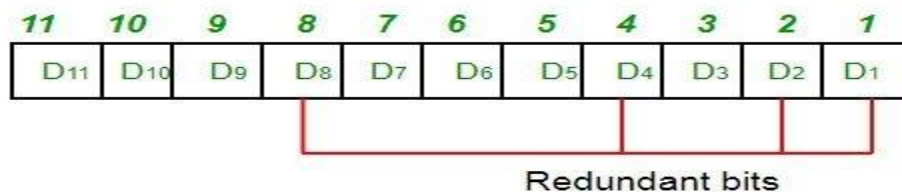
Redundant bits are extra binary bits that are generated and added to the information-carrying bits of data transfer to ensure that no bits are lost during the data transfer.

The number of redundant bits can be calculated using the following formula: $2^r \geq m + r + 1$, where r = redundant bit, m = data bit

Suppose the number of data bits is 7, then the number of redundant bits can be calculated using:

Position of the redundant bits

- For seven-bit ASCII code, redundancy bits are placed in positions 1, 2, 4 and 8.



Position	R8	R4	R2	R1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1

R1 -> 1,3,5,7,9,11
R2 -> 2,3,6,7,10,11
R -> 4,5,6,7
R -> 8,9,10,11

Example

- Suppose the data to be transmitted is 1011001, the bits will be placed as follows:

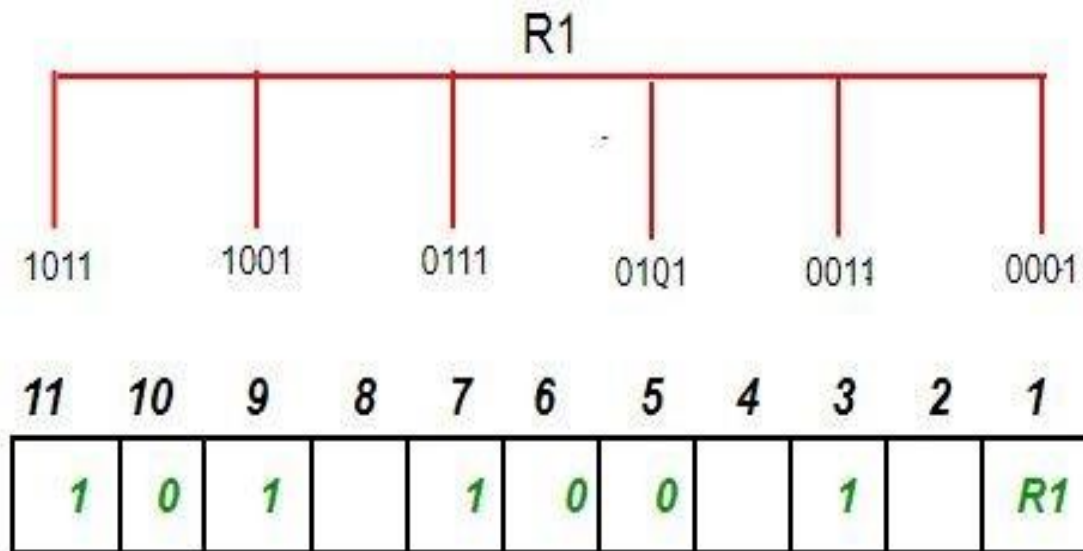
11	10	9	8	7	6	5	4	3	2	1
1	0	1	R8	1	0	0	R4	1	R2	R1

Total number of data bits 'm' = 7

- Number of redundant bits r: $2^r \geq m + r + 1$
 - $2^r \geq 7 + r + 1$
 - $2^4 \geq 7 + 4 + 1 = 12$ (Satisfied)
- Therefore, r=4 and total number of bits = $m+r = 7+4=11$;

Determining the Parity bits:

1. Determining R1 bit: (R1: bits 1, 3, 5, 7, 9, 11)

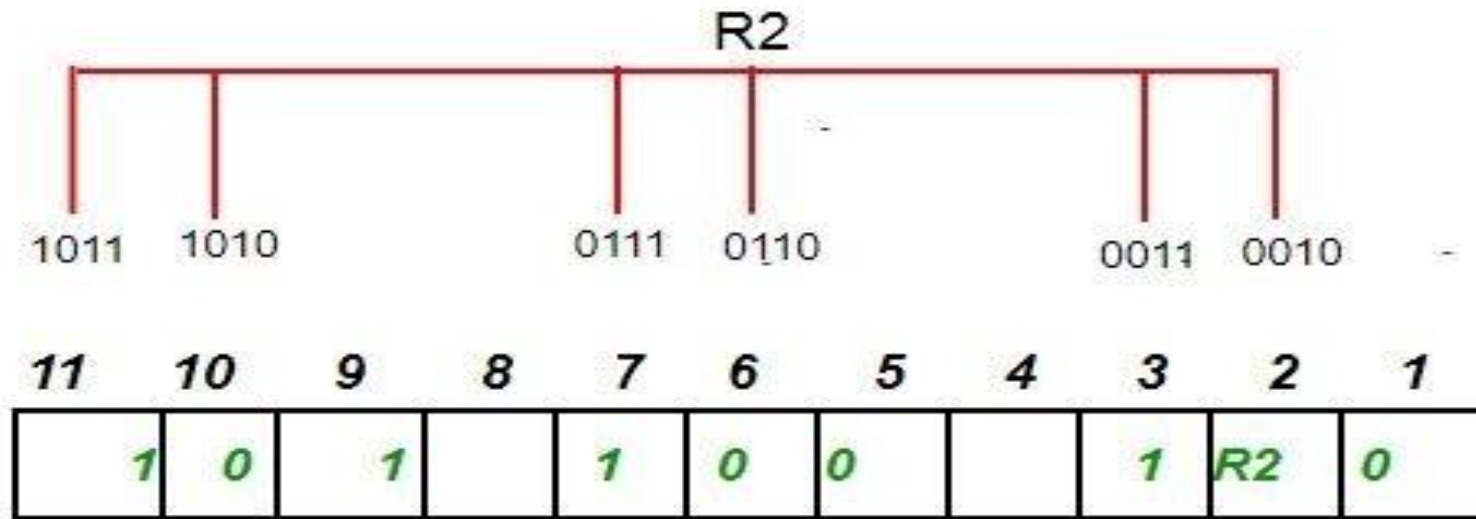


Determining the Parity bits:

To find the redundant bit R1, we check for even parity.

Since the total number of 1's in all the bit positions corresponding to R1 is an even number, the value of R1 = 0

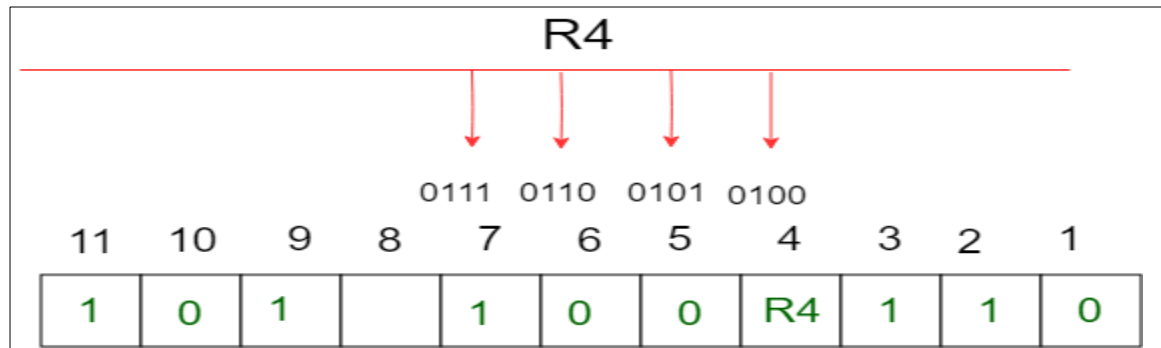
2. Determining R2 bit: (R2: bits 2,3,6,7,10,11)



Determining the Parity bits:

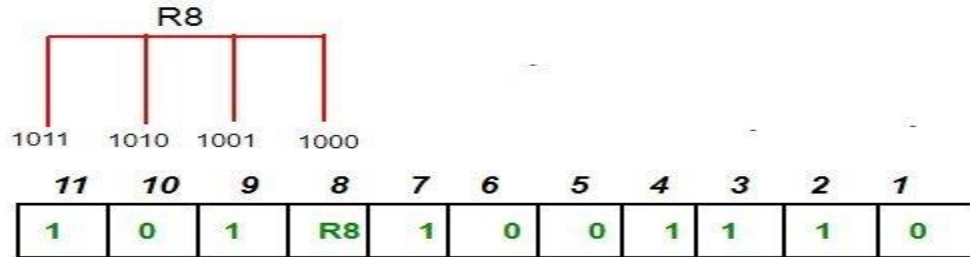
- To find the redundant bit R_2 , we check for even parity.
- Since the total number of 1's in all the bit positions corresponding to R_2 is odd the value of $R_2 = 1$

3. Determining R4 bit: (R4: bits 4, 5, 6, 7)



- To find the redundant bit R4, we check for even parity. Since the total number of 1's in all the bit positions corresponding to R4 is odd the value of $R4 = 1$.

4. Determining R8 bit: (R8: bit 8,9,10,11)

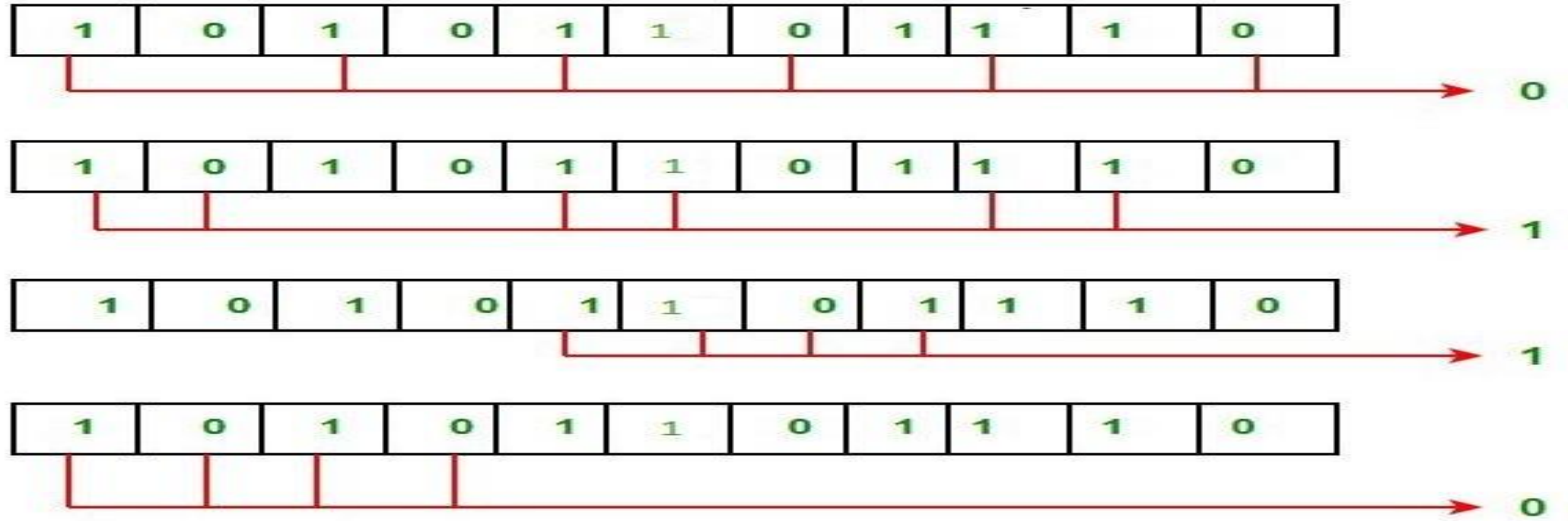


To find the redundant bit R8, we check for even parity. Since the total number of 1's in all the bit positions corresponding to R8 is an even number the value of R8=0

- Thus, the data transferred is

11	10	9	8	7	6	5	4	3	2	1
1	0	1	0	1	0	0	1	1	1	0

- Suppose in the above example the 6th bit is changed from 0 to 1 during data transmission, then it gives new parity values in the binary number:



- The bits give the binary number as 0110 whose decimal representation is 6.
- Thus, the bit 6 contains an error. To correct the error the 6th bit is changed from 1 to 0.

3.4 High-Level Data Link Control (HDLC) & Point-to-Point Protocol (PPP)

1. High-Level Data Link Control (HDLC):

It is a bit-oriented protocol for communication over point-to-point and multipoint links developed by the ISO.

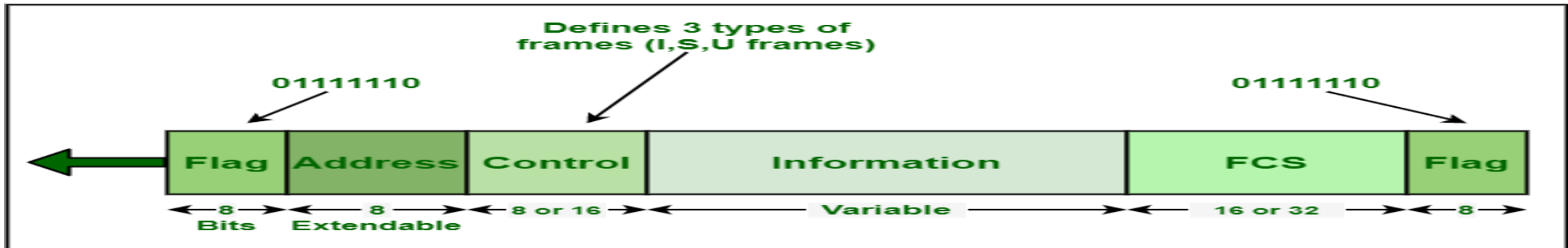
HDLC was developed from the Synchronous Data Link Control (SDLC) standard.

It provides both connection-oriented and connectionless service between two points.

It defines a layer 2 framing structure that allows for flow control and error control through the use of acknowledgments.

It uses a frame delimiter, or flag, to mark the beginning and the end of each frame.

HDLC Framing



Basic Frame Structure

The structure varies according to the type of frame. The fields of an HDLC frame are –

Flag field– It is an 8-bit sequence that marks the beginning and the end of the frame. The bit pattern of the flag is 01111110.

Address field – It contains the address of the receiver. If the frame is sent by the primary station, it contains the address of the secondary station, and vice versa. The address field may be from 1 byte to several bytes.

HDLC Framing

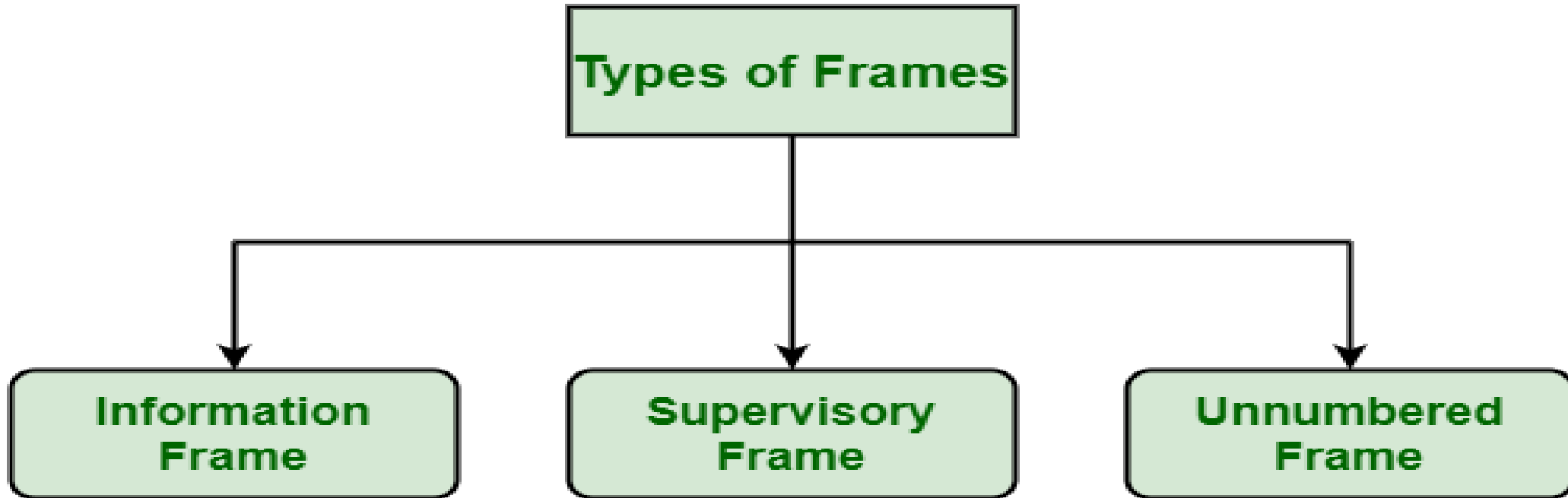
Control field – It is 1 or 2 bytes containing flow and error control information.

Payload field– This carries the data from the network layer.

Frame check Sequence(FCS field)– It is a 2-byte or 4-byte frame check sequence for error detection.

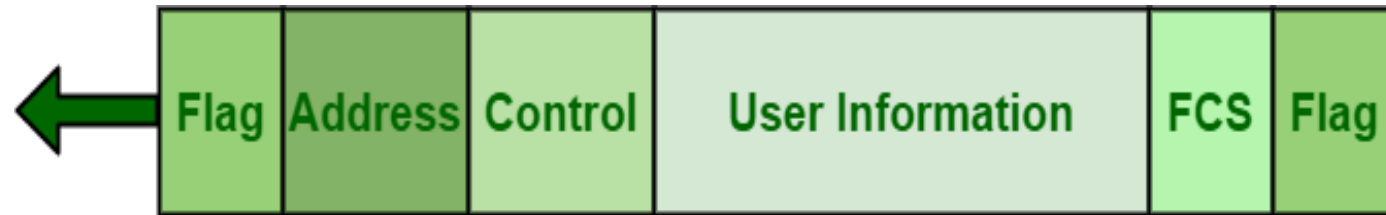
The standard code used is CRC (cyclic redundancy code)
The frame type is determined by the frame's control field.

HDLC defines three types of frames:-



Information frames

1. I-frame: I-frames or Information frames carry user data from the network layer. They also include flow and error-control information piggybacked on user data. The first bit of the control field of the I-frame is 0.



I-frame

Supervisory frames

2.S-frame: S-frames, or Supervisory frames, do not contain an information field. They are used for flow and error control when piggybacking is not required. The first two bits of the control field of the S-frame is 10.



S-frame

Un-numbered frames

3. U-frame: U-frames or unnumbered frames are used for miscellaneous functions, like link management. It may contain an information field, if required. The first two bits of the control field of the U-frame is 11.



U-frame

Point - to - Point Protocol (PPP):

Point - to - Point Protocol (PPP) is a communication protocol of the data link layer that is used to transmit data between two directly connected (point-to-point) computers.

It is a byte - oriented protocol that is widely used in broadband communications, having heavy loads and high speeds.

Since it is a data link layer protocol, data is transmitted in frames.

Many Internet users who need to connect their home computer to the server of an Internet service provider use PPP.

A point-to-point link protocol is required to control and manage the transfer of data.

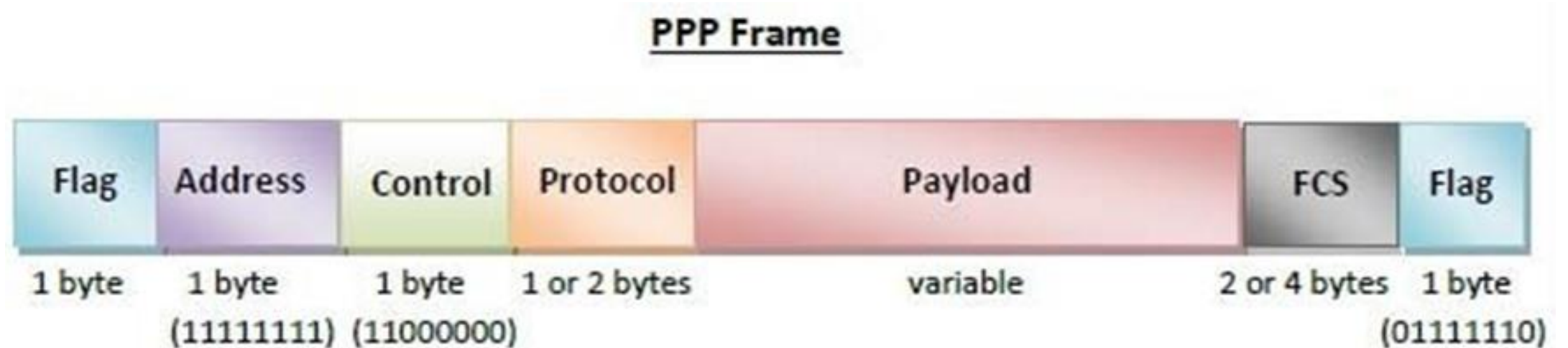
Point - to - Point Protocol (PPP):

PPP provides the format of the frame to be exchanged between devices.

Show two devices negotiate the establishment of the link and the exchange of data, how network layer data are encapsulated in the data link frame, how two devices can authenticate each other, multiple network layer services connection over multiple links, and network address configuration

- But, several services are missing for simplicity: no flow control, simple error control (detection and discard)

Point - to - Point Protocol (PPP):



PPP Framing:

Flag– marks the beginning and the end of the frame. The bit pattern of the flag is 01111110.

Address-it is set to 11111111 in case of broadcast.

Control – set to a constant value of 11000000.(No need because PPP has no flow control and limited error control)

Protocol – 1 or 2 bytes that identify the protocol of data

encapsulated in the payload field of the frame.

Payload – This carries the data from the network layer. The maximum length of the payload field is 1500 bytes

3.5 Channel Allocation Problem

Channel allocation is the process of dividing a single channel and assigning it to multiple users to carry out user-specific tasks.

If there are N users and the channel is divided into N equal-sized sub-channels, each user is assigned one portion.

3.5 Channel Allocation Problem

If the number of users is small and doesn't vary over time, then Frequency Division Multiplexing can be used, as it is a simple and efficient channel bandwidth allocation technique.

The channel allocation problem can be solved by two schemes: Static Channel Allocation and Dynamic Channel Allocation.

a) Static Channel Allocation:

In a static channel allocation scheme, a fixed portion of the frequency channel is allotted to each user.

For N competing users, the bandwidth is divided into N channels using frequency division multiplexing (FDM), and each portion is assigned to one user.

This scheme is also referred to as fixed channel allocation or fixed channel assignment.

In this allocation scheme, there is no interference between the users since each user is assigned a fixed channel.

a) Static Channel Allocation:

- However, it is not suitable in the case of a large number of users with a large variable bandwidth required.

b) Dynamic Channel Allocation:

In a dynamic channel allocation scheme, frequency bands are not permanently assigned to the users.

Instead, channels are allotted to users dynamically as needed, from a central pool.

The allocation is done considering a number of parameters so that transmission interference is minimized.

This allocation scheme optimizes bandwidth usage and results in faster transmissions.

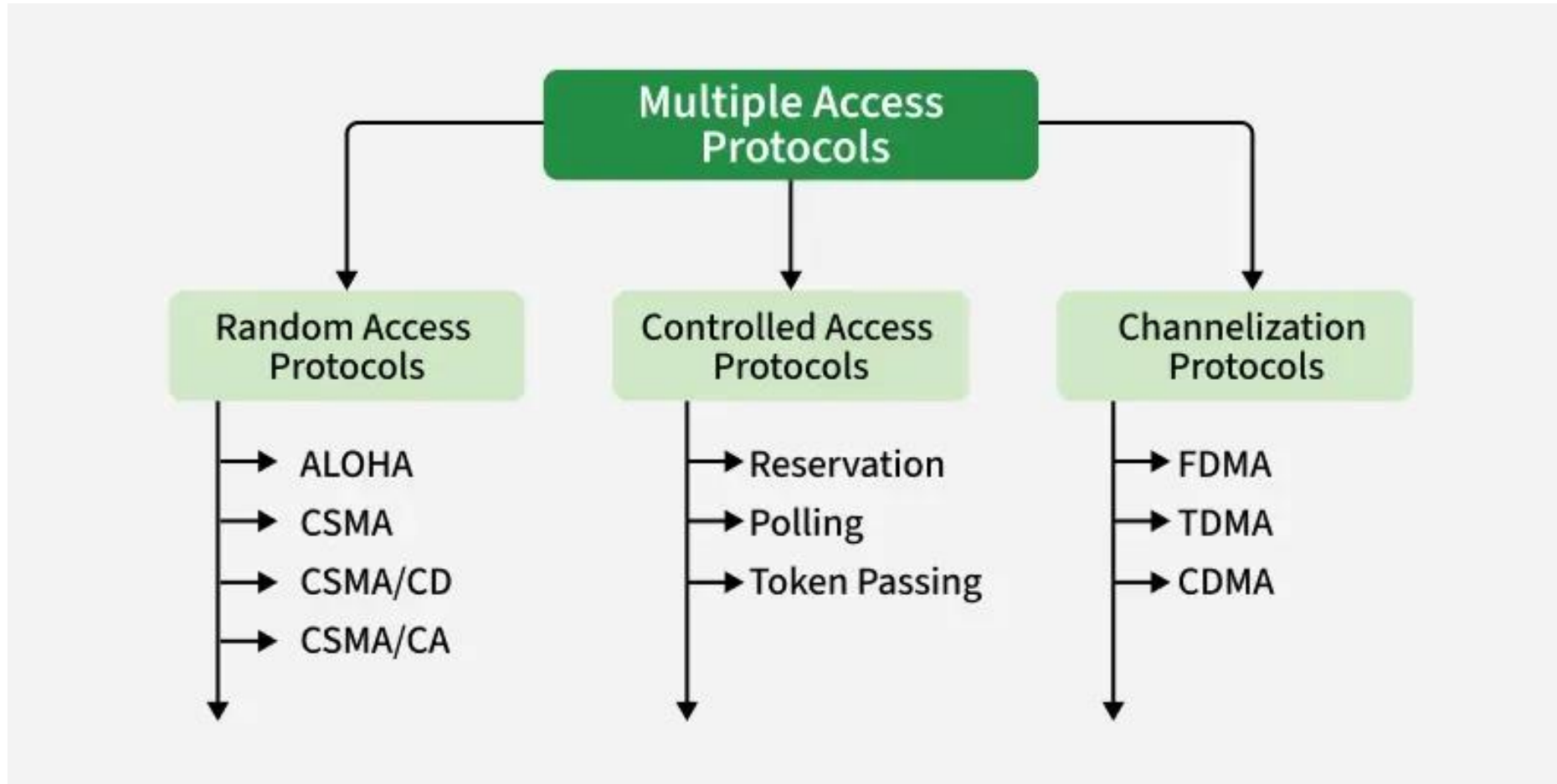
3.6 Multiple Access Protocol

Multiple Access Protocols are a set of rules used in computer networks to control how multiple devices share and access a common communication channel.

Since simultaneous transmissions by multiple devices can lead to data collisions and loss.

These protocols ensure orderly, fair, and efficient use of the shared medium by coordinating when and how each device is allowed to transmit data.

Categories of Multiple Access Protocols



Random access Protocol

In this approach, all stations have equal priority.

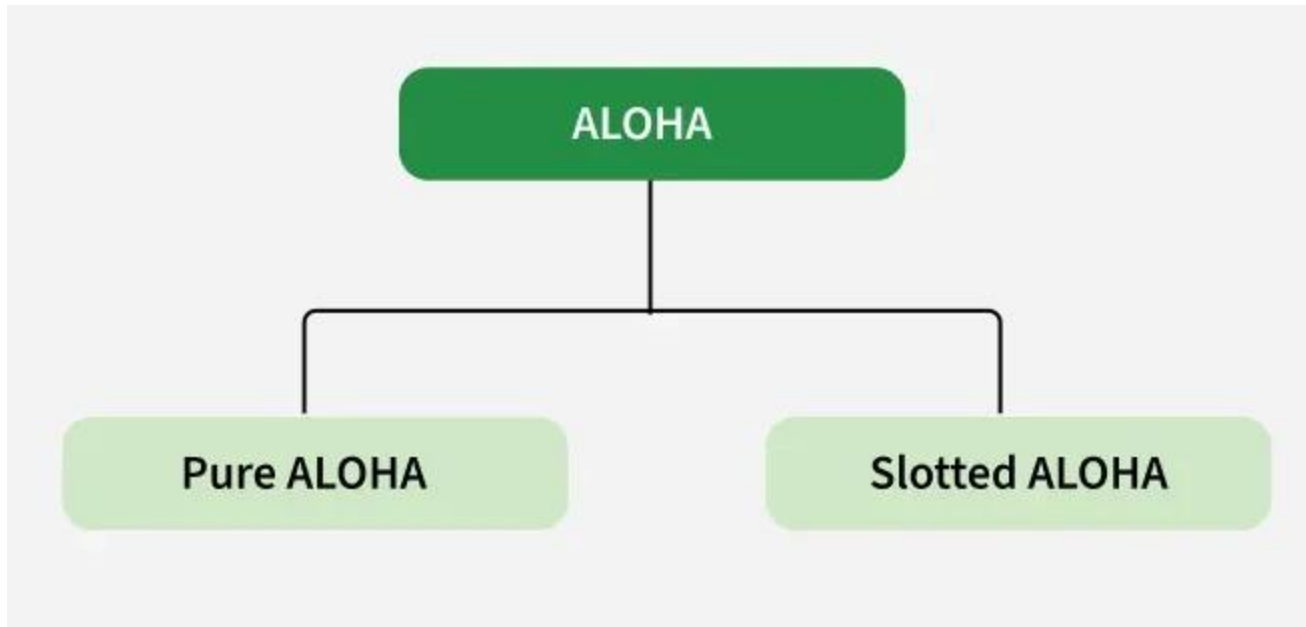
Transmission decisions are based on the state of the channel (idle or busy).

Below are its Features:

- No fixed order of transmission.
- No fixed time slot assignment.

a) ALOHA:

- Designed for wireless LANs and shared media.
- Multiple stations can transmit simultaneously, leading to collisions.



Pure Aloha:

Pure Aloha:

When a station sends data, it waits for an acknowledgement.

If the acknowledgement doesn't come within the allotted time, then the station waits for a random amount of time called back-off time (T_b) and re-sends the data.

Since different stations wait for different amounts of time, the probability of further collision decreases.

Pure Aloha:

Vulnerable Time = $2 * \text{Frame transmission time}(T_{fr})$

S = Throughput (successful transmissions)

G = Offered load (number of transmission attempts per frame time)

Throughput = $G * \exp^{-2 * G}$ where G is the number of transmission attempts in the T_{fr} slot

Maximum throughput = 0.184 for $G=0.5$

i.e, only about 18.4% of the time is used for successful transmissions

Slotted Aloha:

It is similar to pure aloha, except that we divide time into slots and sending of data is allowed only at the beginning of these slots.

If a station misses out the allowed time, it must wait for the next slot.

This reduces the probability of collision.

Slotted Aloha:

Vulnerable Time = Frame transmission time (T_{fr})

Throughput = $G * \exp^{-G}$

Maximum throughput = 0.368 for $G=1$

- i.e; about 36.8% of the time is used for successful transmissions.

b) CSMA:

It is a carrier-sense multiple access/collision detection network protocol to transmit data frames.

It first senses the shared channel before transmitting the frames, and if the channel is idle, it transmits a frame to check whether the transmission was successful.

If the frame is successfully received, the station sends another frame.

If any collision is detected in the CSMA/CD, the Station sends a jam/stop signal to the shared channel to terminate data transmission. After that, it waits for a random time before sending a frame to a channel.

c) CSMA/CD:

Carrier Sense Multiple Access with Collision detection network protocol to transmit data frames.

It first senses the shared channel before transmitting the frames, and if the channel is idle, it transmits a frame to check whether the transmission was successful.

If the frame is successfully received, the station sends another frame.

If any collision is detected in the CSMA/CD, the Station sends a jam/stop signal to the shared channel to terminate data transmission.

After that, it waits for a random time before sending a frame to a channel.

CSMA/CA:

In Carrier Sense Multiple Access collision Avoidance CSMA/CA, as soon as a node is ready to send a packet, it checks to be sure the channel is clear (no other node is transmitting at the time).

If the channel is clear, then the packet is sent. If the channel is not clear, the node waits for a randomly chosen period of time and then checks again to see if the channel is clear.

This period of time is called the backoff factor, and is counted down by a backoff counter.

If the channel is clear when the backoff counter reaches zero, the node transmits the packet. If the channel is not clear when the backoff counter reaches zero, the backoff factor is set again, and the process is repeated.

These are three types of strategies to avoid collision:

a) Inter Frame Space (IFS) - Station waits for medium to become idle and if found idle it does not immediately send data rather it waits for a period of time called IFS. After this time, it again checks the medium for being idle. IFS duration depends on the priority of station.

b) Contention Window - It is the amount of time divided into slots. A station which is ready to send frames chooses random number of slots as wait time.

C) Acknowledgment-Sender re-transmits the data if acknowledgment is not received before time-out.

Controlled Access Protocols:

In controlled access, the stations seek information from one another to find which station has the right to send.

It allows only one node to send at a time, to avoid collision of messages on shared medium.

The three controlled-access methods are:

- a) Reservation
- b) Polling
- c) Token Passing

Reservation:

In the reservation method, a station needs to make a reservation before sending data.

The timeline has two kinds of periods:

Reservation interval of fixed time length

Data transmission period of variable frames.

If there are M stations, the reservation interval is divided into M slots, and each station has one slot.

Suppose that station 1 has a frame to send, it transmits 1 bit during slot 1. No other station is allowed to transmit during this slot.

Reservation:

The stations that have reserved their slots transfer their frames in that order.

After the data transmission period, the next reservation interval begins.

Since everyone agrees on who goes next, there will never be any collisions.

b) Polling:

The polling process is similar to the roll call performed in class. Just like the teacher, a controller sends a message to each node in turn.

In this, one acts as a primary station(controller), and the others are secondary stations.

All data exchanges must be made through the controller.

The message sent by the controller includes the address of the node to be granted access.

b) Polling:

Although all nodes receive the message, the addressed one responds to it and sends data, if any.

If there is no data, usually a “poll reject”(NAK) message is sent back.

Problems include the high overhead of the polling messages .

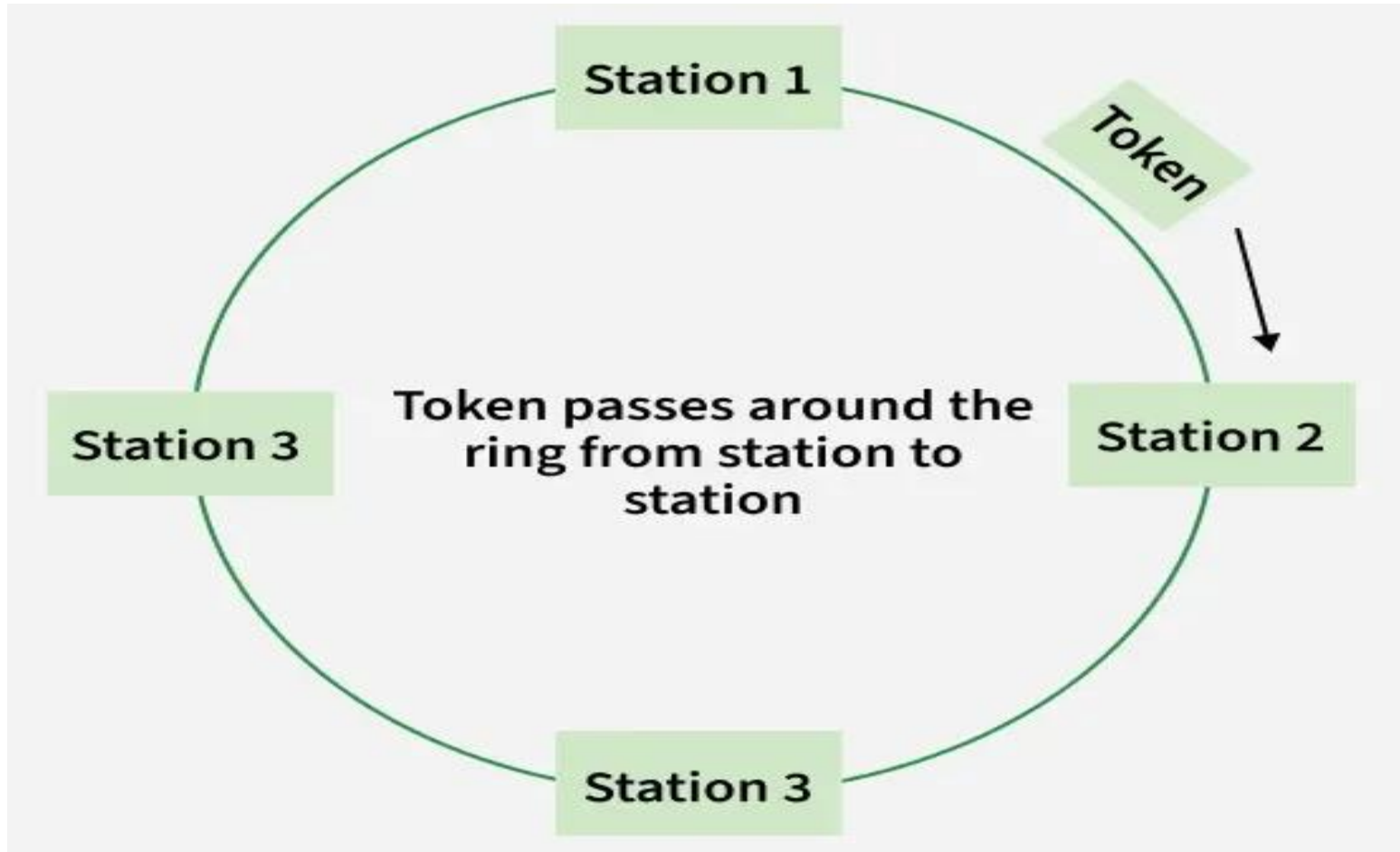
Efficiency

- Let T_{poll} be the time for polling and T_t be the time required for transmission of data. Then,
- Efficiency = $T_t / (T_t + T_{\text{poll}})$

Token Passing:

- In Token Passing, stations are logically connected (ring or bus). A special frame called a token circulates among stations:
- The token represents permission to transmit.
- A station can send data only when it holds the token.
- If no data is ready, the token is simply passed on.
-

Token Passing:



Token Passing:

Common Implementations

- **Token Ring** – token passed in a ring topology.
- **Token Bus** – stations use a bus medium but pass the token in a predefined logical order.
- **Advantages**
 - Excellent performance under high loads.
 - Built-in error recovery and debugging features.
 - Predictable throughput.
- **Disadvantages**
 - Expensive setup and hardware.
 - Token management issues (lost/duplicate tokens, station failures).
 - Complex implementation.

Feature	Reservation	Polling	Token Passing
Access Control	Slot-based reservation	Central controller polls	Token circulation
Efficiency	High (under heavy load)	Moderate (polling overhead)	High (predictable)
Collision Chance	None	None	None
Overhead	Reservation slots	Polling messages	Token management
Best For	Real-time multimedia	Small/medium networks	Large/high-load networks

Channelization Protocols:

In this, the available bandwidth of the link is shared in time, frequency, and code to multiple stations to access channel simultaneously.

- a) Frequency Division Multiple Access(FDMA)
- b) Time Division Multiple Access(TDMA)
- c) Code Division Multiple Access(CDMA)

Frequency Division Multiple Access (FDMA)

The available bandwidth is divided into equal bands so that each station can be allocated its own band.

Guard bands are also added so that no two bands overlap to avoid crosstalk and noise.

Time Division Multiple Access (TDMA)

In this, the bandwidth is shared between multiple stations.

To avoid collision, time is divided into slots and stations are allotted these slots to transmit data.

Code Division Multiple Access (CDMA)

One channel carries all transmissions simultaneously.

There is neither a division of bandwidth nor division of time.

Data from different stations can be transmitted simultaneously in different code languages.

3.7 Wired LAN: Ethernet Standards and FDDI

1. Ethernet Standards:

Ethernet is a set of technologies and protocols that are used primarily in LANs.

Ethernet is classified into two categories:

1. Classic Ethernet and
2. Switched Ethernet.

Ethernet

A Classic Ethernet is the original form of Ethernet that provides data rates between 3 and 10 Mbps.

The varieties are commonly referred to as 10BASE-X.

Here, 10 is the maximum throughput, i.e., 10 Mbps, BASE denotes the use of baseband transmission, and X is the type of medium used.

A Switched Ethernet uses switches to connect to the stations in the LAN

2. Fiber Distributed Data Interface (FDDI):

Fiber Distributed Data Interface (FDDI) is a set of ANSI(American National Standards Institute) and ISO standards for transmission of data in local area network (LAN) over fiber optic cables.

Features

FDDI uses optical fiber as its physical medium.

It operates in the physical and medium access control (MAC layer) of the Open Systems Interconnection (OSI) network model.

It provides a high data rate of 100 Mbps and can support thousands of users.

2. Fiber Distributed Data Interface (FDDI):

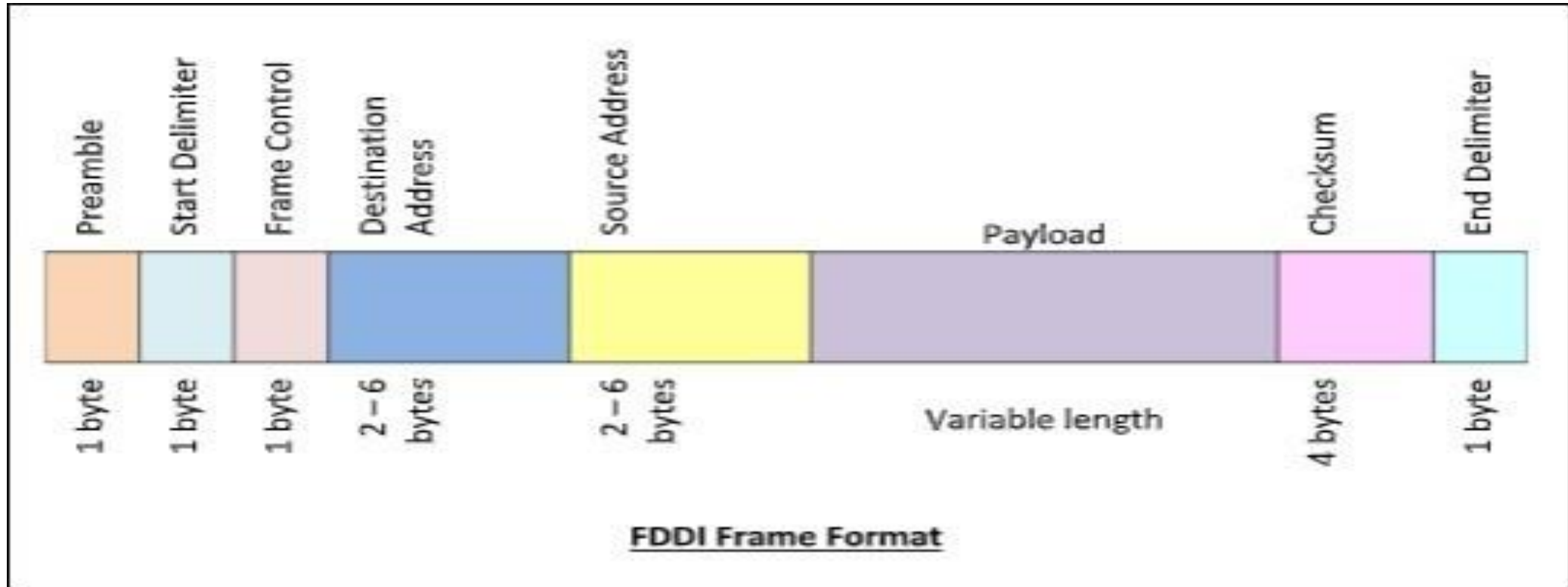
It contains two token rings, a primary ring for data and token transmission and a secondary ring that provides backup if the primary ring fails.

FDDI technology can also be used as a backbone for a wide area network (WAN).

- Frame Format

The frame format of FDDI is as shown in the following diagram –

2. Fiber Distributed Data Interface (FDDI):



The fields of an FDDI frame are –

Preamble: 1 byte for synchronization.

Start Delimiter: 1 byte that marks the beginning of the frame.

Frame Control: 1 byte that specifies whether this is a data frame or control frame.

Destination Address: 2-6 bytes that specifies address of destination station.

The fields of an FDDI frame are –

Source Address: 2-6 bytes that specifies address of source station.

Payload: A variable length field that carries the data from the network layer.

Checksum: 4 bytes frame check sequence for error detection.

End Delimiter: 1 byte that marks the end of the

3.8 Wireless LAN: IEEE 802.11X and Bluetooth Standards

1. IEEE 802.11 Standards:

Wireless LANs (WLANs) are wireless computer networks that use high-frequency radio waves instead of cables for connecting the devices within a limited area, forming a LAN (Local Area Network).

Users connected by wireless LANs can move around within this limited area, such as a home, school, campus, office building, railway platform, etc.

Most WLANs are based upon the standard IEEE

Common Versions

Standard	Frequency	Speed	Description
802.11a	5 GHz	54 Mbps	Less interference
802.11b	2.4 GHz	11 Mbps	Older, slower
802.11g	2.4 GHz	54 Mbps	Improved version
802.11n	2.4/5 GHz	600 Mbps	Uses MIMO
802.11ac	5 GHz	>1 Gbps	High speed
802.11ax (Wi-Fi 6)	2.4/5 GHz	Very high	Latest, efficient

Advantages of WLANs:

They provide clutter-free homes, offices, and other networked places.

The LANs are scalable in nature. The system is portable within the network coverage.

Installation and setup are much easier counterparts. The equipment and setup costs are reduced.

Disadvantages of WLANs:

Since radio waves are used for communications, the signals are noisier with more interference from nearby systems.

Greater care is needed for encrypting information.

WLANs are slower than wired LANs.

Bluetooth Standards:

It is a Wireless Personal Area Network (WPAN) technology and is used for exchanging data over smaller distances.

It operates in the unlicensed, industrial, scientific, and medical (ISM) band at 2.4 GHz to 2.485 GHz.

Bluetooth ranges up to 10 meters. It provides data rates up to 1 Mbps or 3 Mbps, depending on the version.

The spreading technique that it uses is FHSS (Frequency Hopping Spread Spectrum).

A Bluetooth network is called a piconet. collection of interconnected piconets is a scatternet.

Bluetooth Architecture:

The architecture of Bluetooth defines two types of networks:

- a) Piconet
- b) Scatternet

a)Piconet:

A Bluetooth network is called a piconet or a small net.

A piconet can have up to eight stations, one of which is called the primary; the rest are called secondaries.

The secondary stations are synchronized with the primary station.

Communication between the primary and secondary stations can be one-to-one or one-to-many.

b)Scatternet:

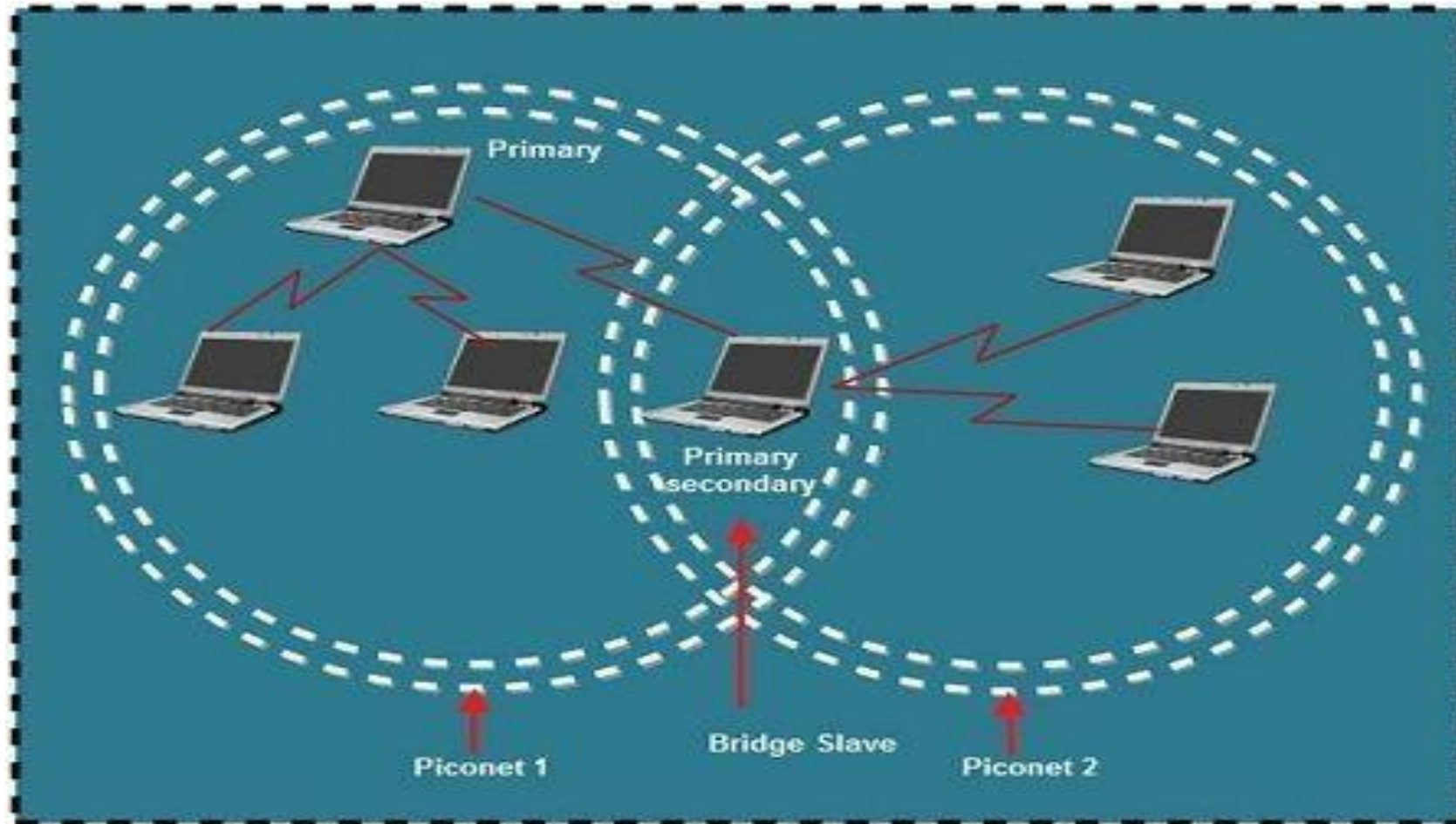
Piconets can be combined to form what is called a scatternet.

A secondary station in one piconet can be the primary in another piconet.

This station can receive messages from the primary in the first piconet (as a secondary) and, acting as a primary, deliver them to secondaries in the second piconet.

A station can be a member of two piconets.

Scatternet:



3.9 Token Bus, Token Ring and Virtual LAN

1. Token Bus:

Token Bus (IEEE 802.4) is a popular standard for the token passing LANs.

In a token bus LAN, the physical media is a bus and a logical ring is created using coaxial cable.

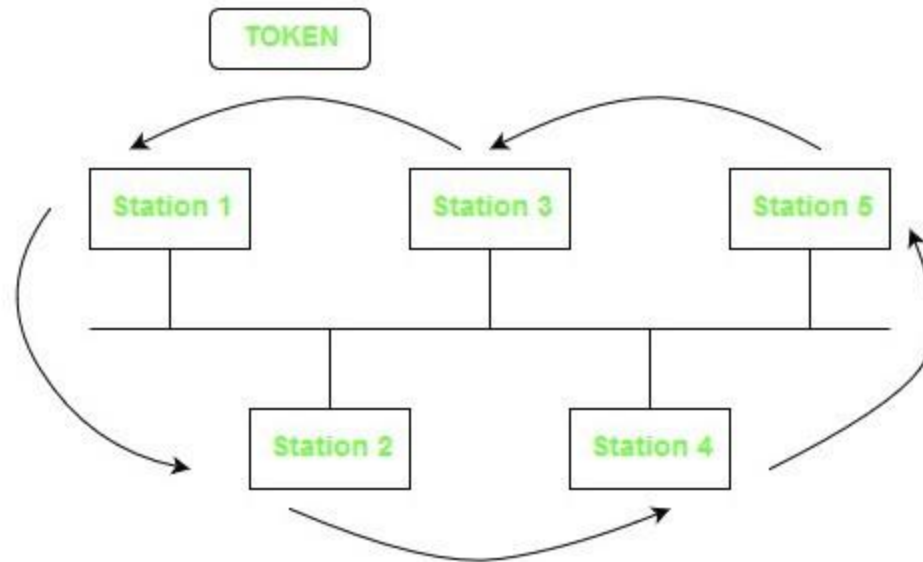
The token is passed from one user to another in a sequence (clockwise or anticlockwise).

Each station knows the address of the station to its “left” and “right” as per the sequence in the logical ring.

A station can only transmit data when it has the token.

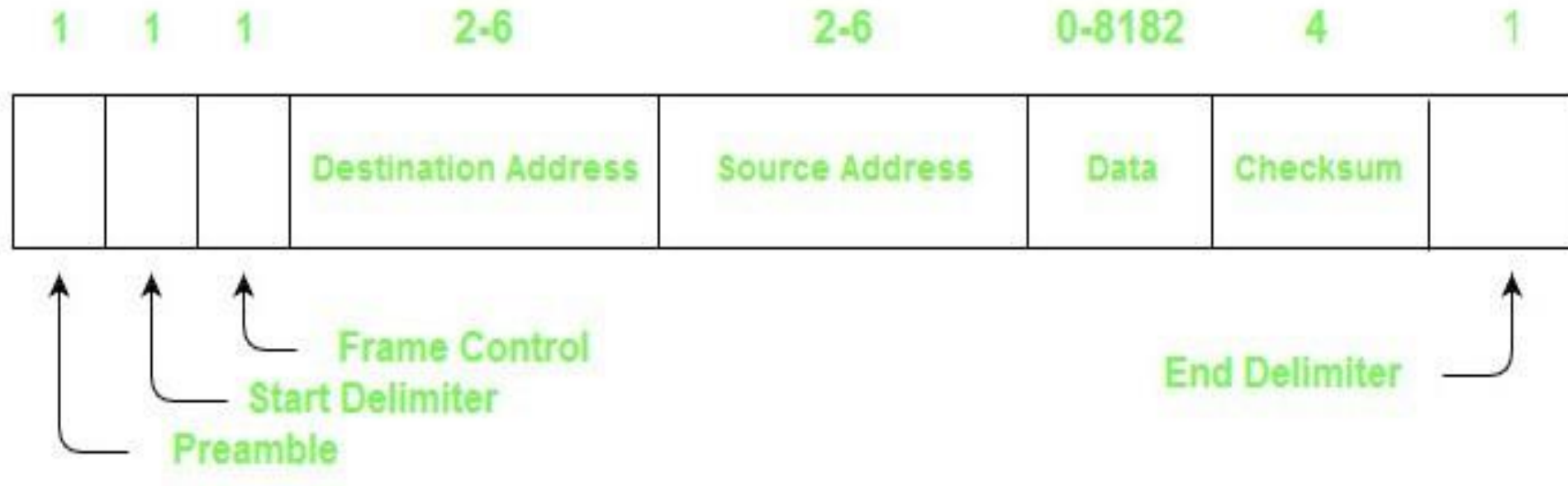
3.9 Token Bus, Token Ring and Virtual LAN

- The above diagram shows a logical ring formed in a bus based token passing LAN. The logical ring is shown with the arrows.



Frame Format:

- The various fields of the frame format are:



Frame Format:

Preamble-It is used for bit synchronization. It is a 1-byte field.

Start Delimiter-These bits mark the beginning of the frame. It is a 1-byte field.

Frame Control - This field specifies the type of frame: data frame and control frames. It is a 1-byte field.

Destination Address - This field contains the destination address. It is a 2 to 6-byte field.

Source Address - This field contains the source address. It is a 2 to 6-byte field.

Data - If 2-byte addresses are used, then the field may be up to 8182 bytes and 8174 bytes in the case of 6 byte Addresses.

Checksum - This field contains the checksum bits, which are used to detect errors in the transmitted data. It is 4 bytes field.

End Delimiter - This field marks the end of frame. It is 1

- **Maximum frame size = 8190 bytes**

Frame Structure:

- **Destination Address (DA) = 2 bytes**
- **Source Address (SA) = 2 bytes**
- **FCS (error checking) = 4 bytes**

2+2+4=9 bytes

Token Ring:

Token ring (IEEE 802.5) is a communication protocol in a local area network (LAN) where all stations are connected in a ring topology and pass tokens for channel acquisition.

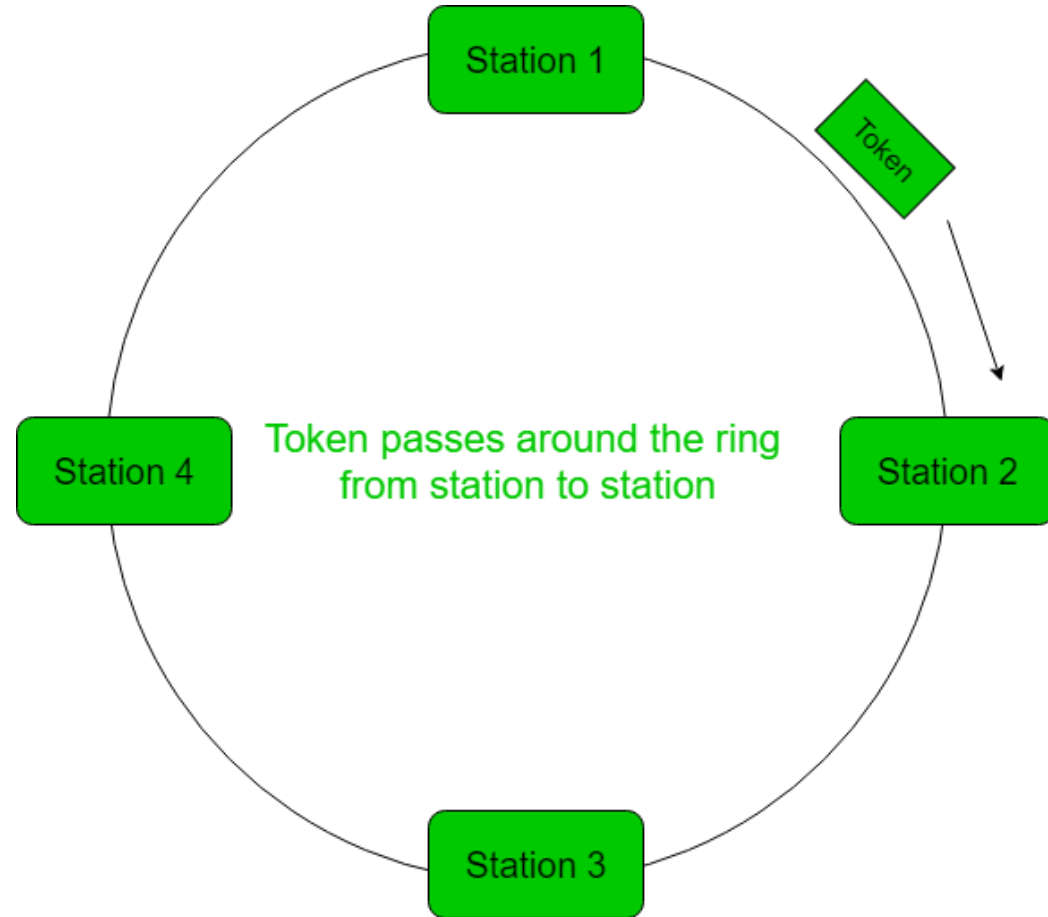
A token is a special frame of 3 bytes that circulates along the ring of stations.

A station can send data frames only if it holds a token.

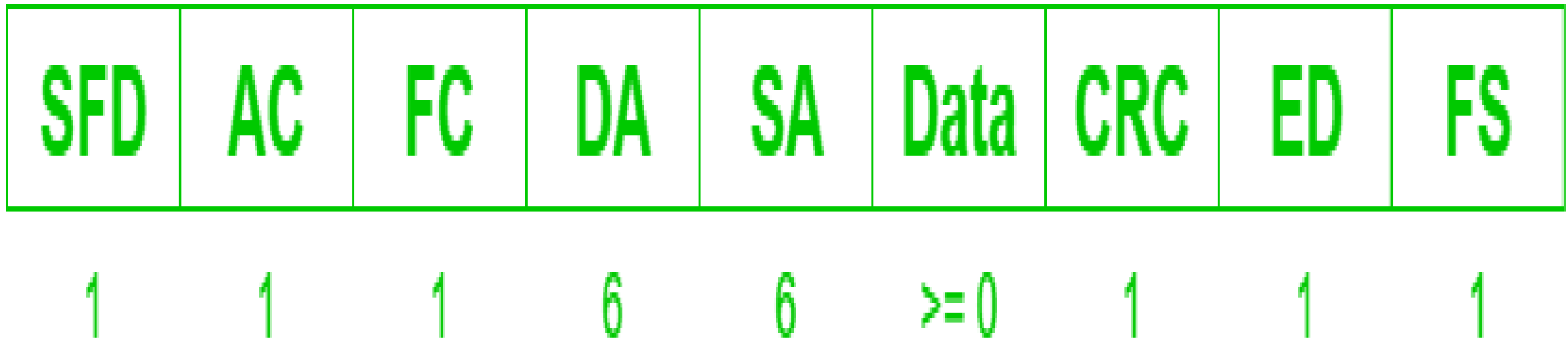
If a station has a frame to transmit when it receives a token, it sends the frame and then passes the token to the next station; otherwise, it simply passes the token to the next station.

The data flow is unidirectional in the direction of the token passing.

Token Ring:



Data Frame



Token Ring:

Start frame delimiter (SFD) and (ED): Used to mark the beginning and end of the frame.

Access control (AC): It is used to represent the 3 byte token.

Frame control (FC): This field specifies the type of frame - data frame and control frames.

Source address and destination address: These fields contain source address and destination address.

Token Ring:

Data: A variable length field that carries the data from the network layer.

Cyclic redundancy check (CRC): CRC is used to check for errors in the frame, i.e., whether the frame is corrupted or not. If the frame is corrupted, then its discarded.

Frame status (FS) - It is a 1-byte field terminating a data frame.

Virtual LAN:

A VLAN (Virtual Local Area Network) is a technology that partitions a single physical network switch into multiple, isolated, and logical networks.

It enhances security and performance by restricting traffic between groups, even if they share the same hardware.

Common examples include isolating IoT devices, separating department traffic (e.g., HR from Engineering), and segmenting guest Wi-Fi from secure office networks



Features:

Virtual LANs offer a structure for creating groups of devices, even if their networks are different.

It can reduce congestion by sharing traffic as an individual VLAN works as a separate LAN.

Advantages:

It can provide flexibility.

It decreases the latency and traffic load on the network and the network devices, offering increased performance.

Disadvantages:

It is more effective at controlling latency than a WAN but less efficient than a LAN.